IEEC
Institut d'Estudis Espacials de Catalunya

SPCOMNAV

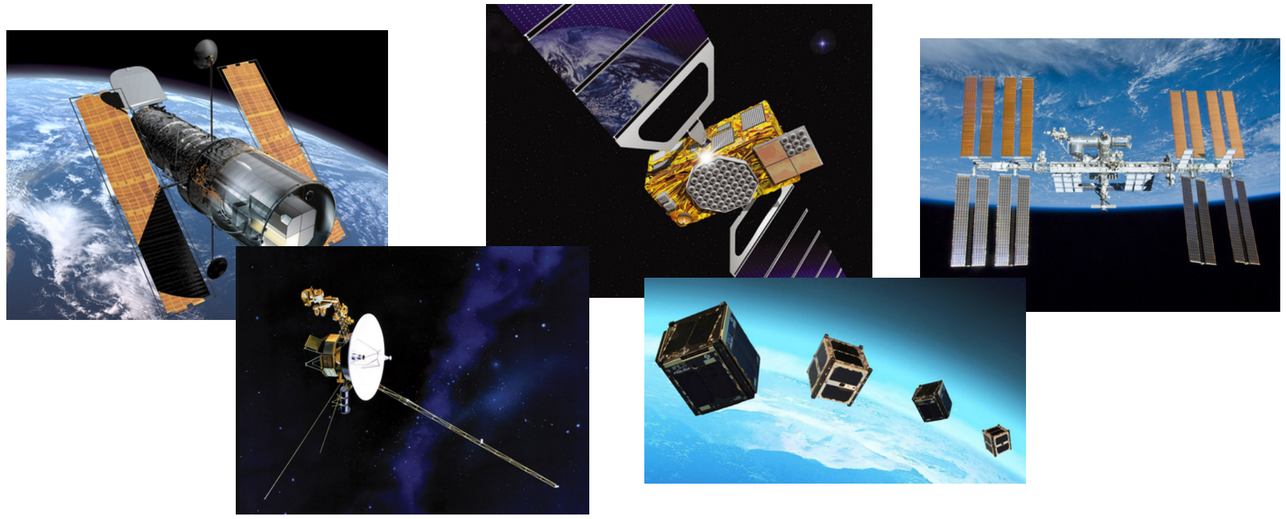
UAB
Universitat Autònoma de Barcelona

Desenvolupament d'una estació de seguiment de megaconstel·lacions de satèl·lits LEO mitjançant un dispositiu RFSoC programable d'alta velocitat

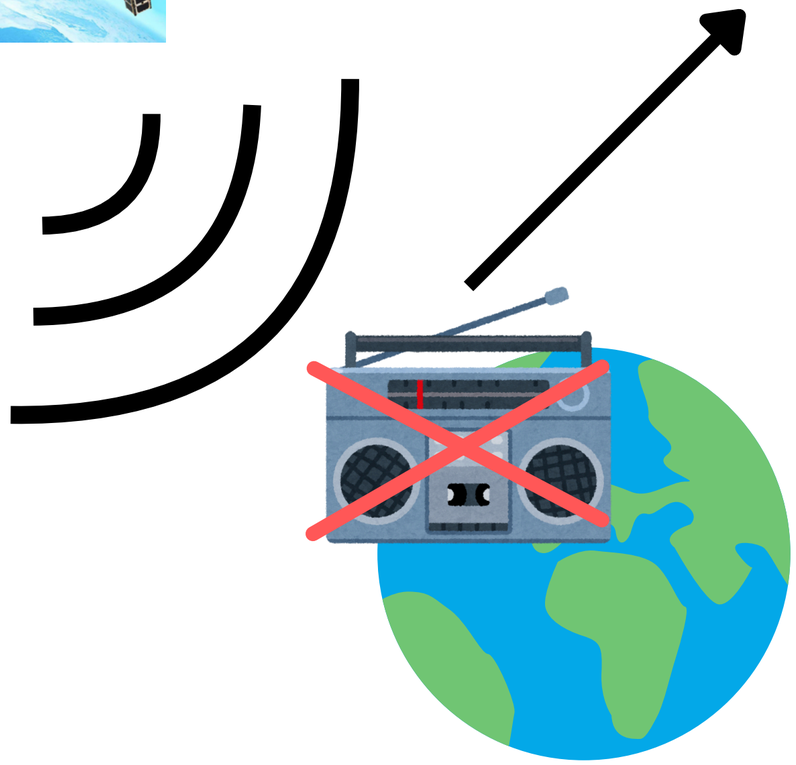
Jan Moran Ricardo

Director: José A. López Salcedo

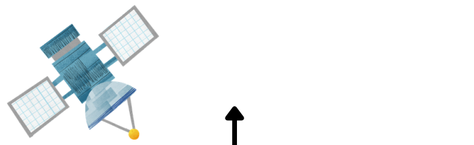
Treball de final de grau - Enginyeria Informàtica

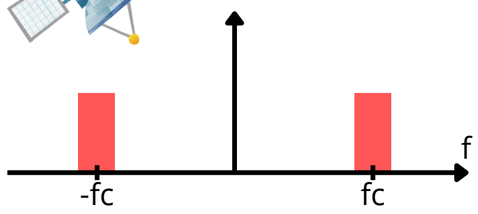


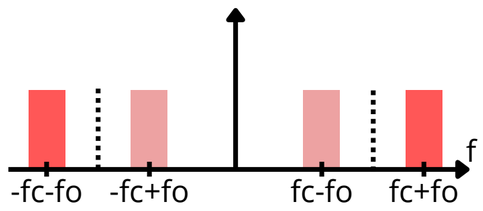
Software Defined Radios (SDRs)

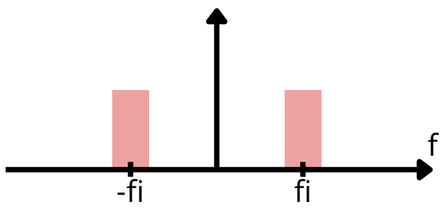


Arquitectura Superheterodina (simplificada)









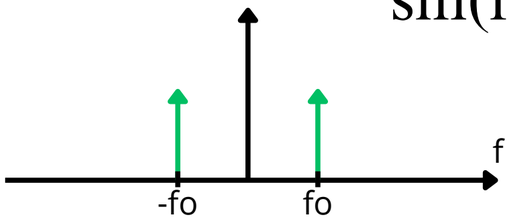
X(fc)

Filtre Fi

Y(fi)

ADC*

sin(fo)



fc = freqüència central
fo = freqüència oscil·lador local
fi = freqüència intermitja

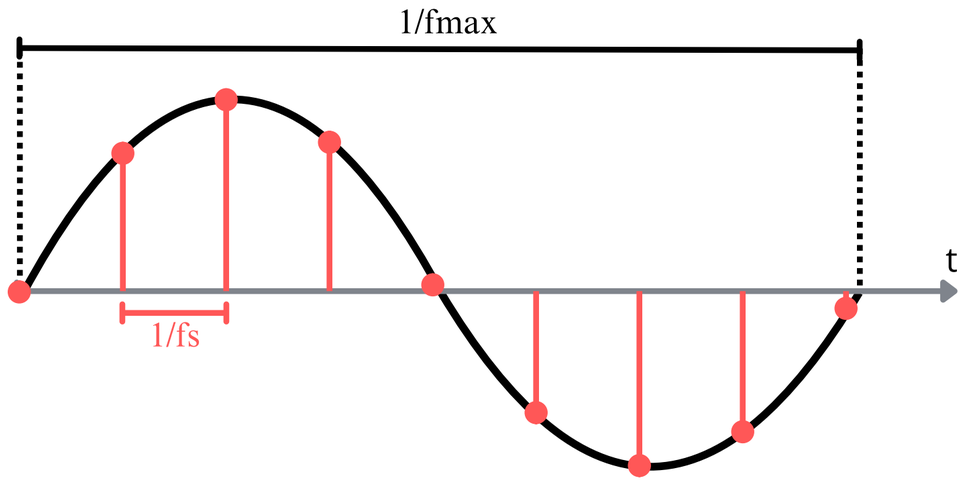
$$fi = fc\text{-}fo$$

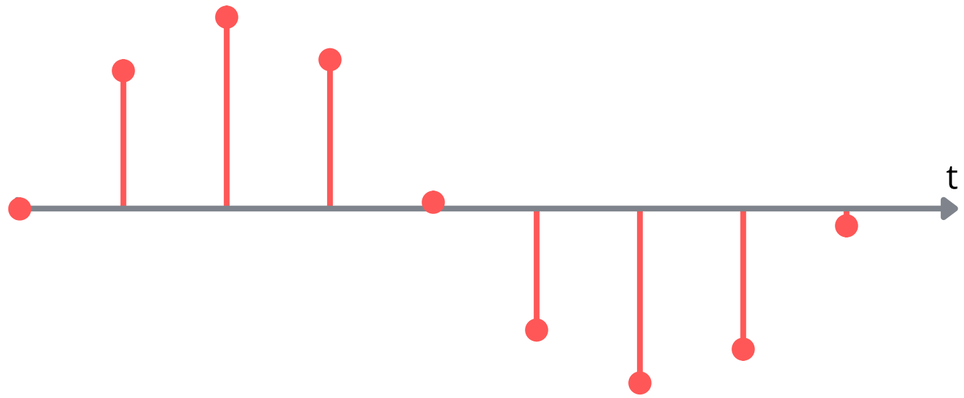
NA: També existeix l'arquitectura **zero-if**, però els principis són els mateixos

*Poso ADC per sibolitzar el canvi de analògic a digital, aquest component serà diferent depenent de l'aplicació.

USRP X440: Mostreig Directe

1. **Estudiar i comprendre** la USRP X440 i l'arquitectura de Mostreig Directe.
2. **Desenvolupar** una **UI i** un **programari** que permeti a altres membres del departament fer us del dispositiu,
3. **Realitzar proves de captura de senyal** que validin els coneixements i el programari.

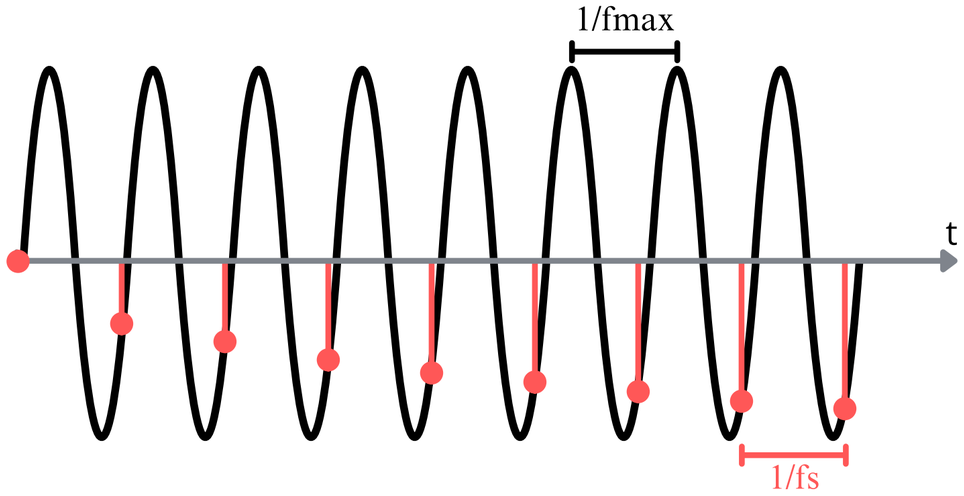


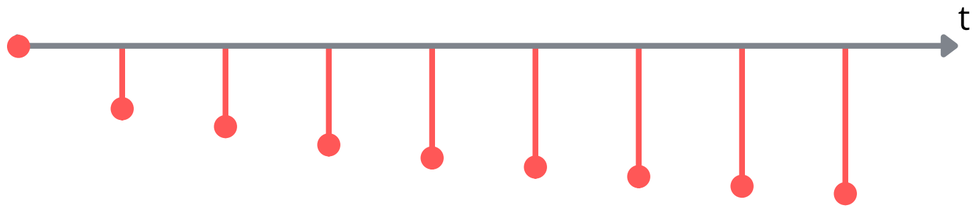


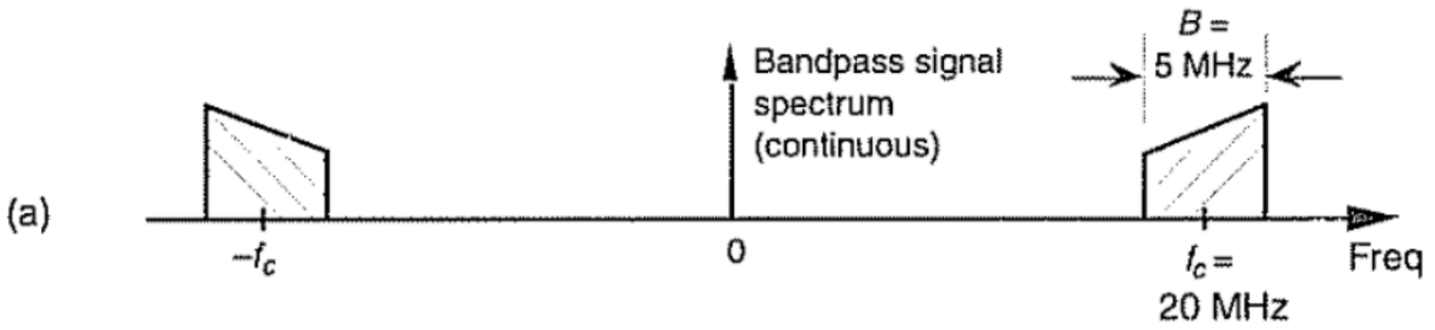
Teorema de Nyquist-Shannon

$$f_s > 2 \cdot f_{max}$$

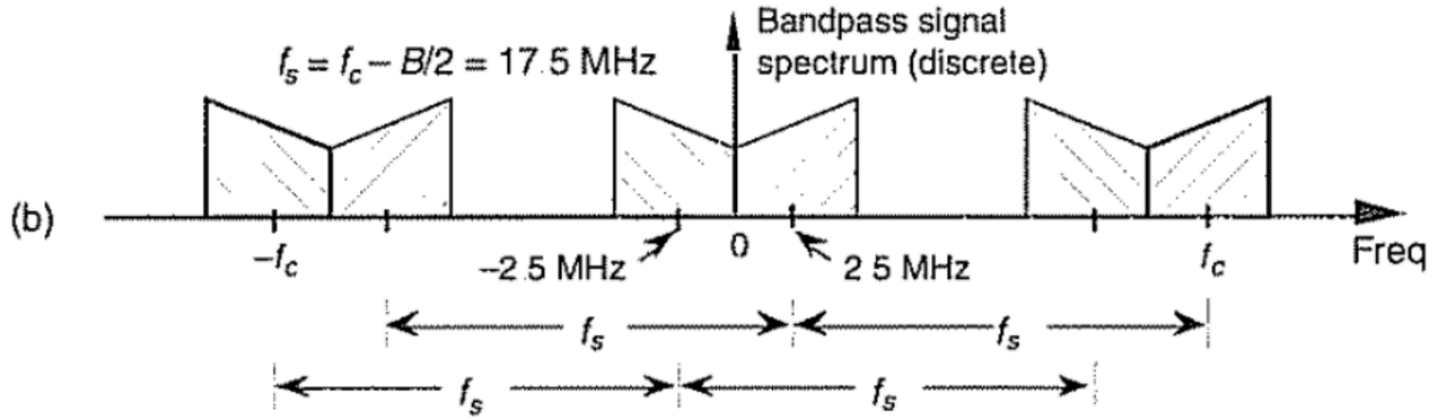
*suposa fmin=0





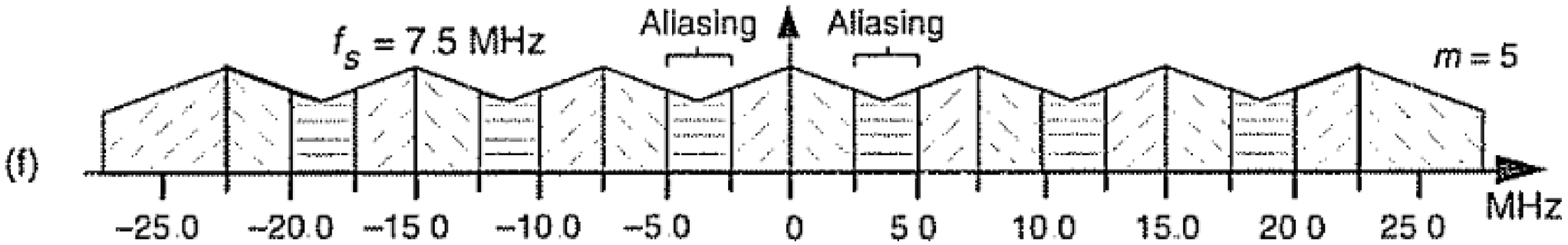


Segons Nyquist: $f_s > 45MHz$



Font: Richard G. Lyons. Understanding Digital Signal Processing. 3rd. Upper Saddle River, NJ: Prentice Hall, 2010. ISBN: 978-0137027415.

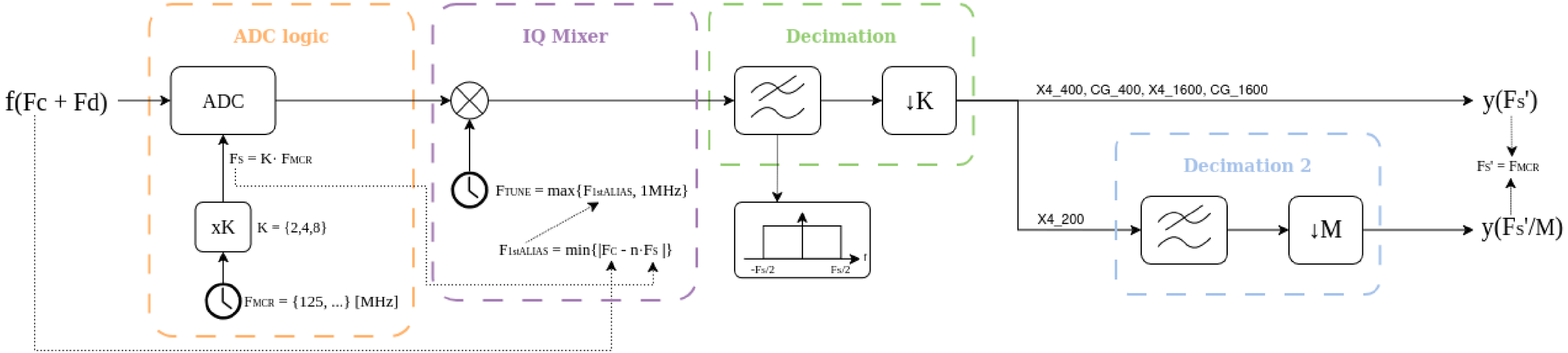
Problema: Alàsing

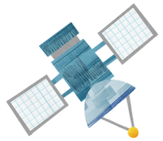


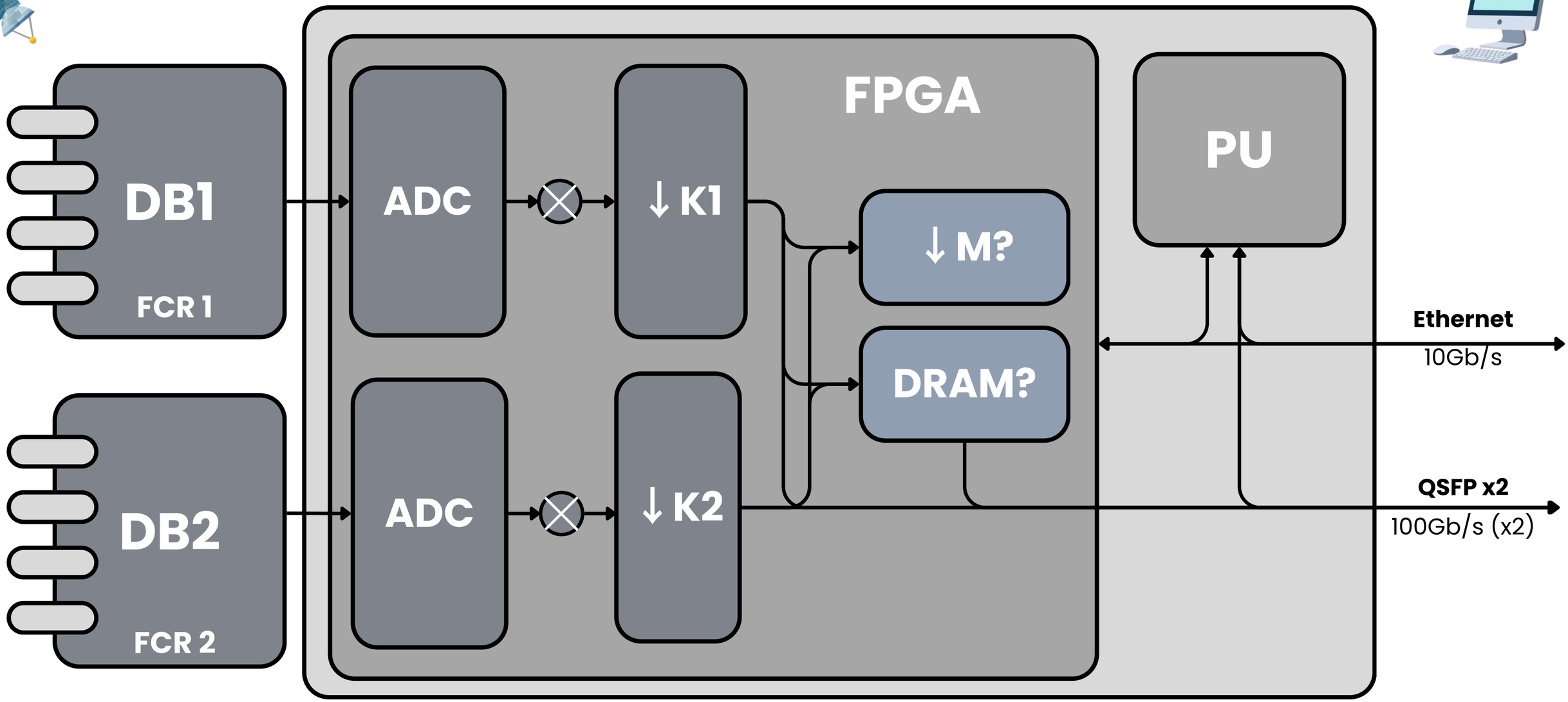
$$\frac{2f_c - B}{m} \geq f_s \geq \frac{2f_c + B}{m}$$

m = Nombre de rèpliques.

Font: Richard G. Lyons. Understanding Digital Signal Processing. 3rd. Upper Saddle River, NJ: Prentice Hall, 2010. ISBN: 978-0137027415.


ADC logic
IQ Mixer
Decimation
f(Fc + Fd)
ADC
Fs = K· FMCR
xK
K = {2,4,8}
FMCR = {125, ...} [MHz]
FTUNE = max{F1stALIAS, 1MHz}
F1stALIAS = min{|Fc - n·Fs |}
↓K
-Fs/2
Fs/2
f
X4_400, CG_400, X4_1600, CG_1600
y(Fs')
Fs' = FMCR
Decimation 2
X4_200
↓M
y(Fs'/M)




DB1
FCR 1
DB2
FCR 2
ADC
ADC
↓ K1
↓ K2
FPGA
↓ M?
DRAM?
PU
Ethernet
10Gb/s
QSFP x2
100Gb/s (x2)

Flavour	Chan.	Bandwidth per Channel	QSFP28 Port 0	QSFP28 Port 1	DDC DUC	DRAM
X4_200	8	200 MHz	4x 10 GbE	Unused	Yes	No
X4_400	8	400 MHz	4x 10 GbE	Unused	No	Yes
CG_400	8	400 MHz	100 GbE	100 GbE	No	No
X4_1600	2	1600 MHz	4x 10 GbE	Unused	No	Yes
CG_1600	2	1600 MHz	100 GbE	100 GbE	No	No

USRP X440 - wizard

F_c: 2 M G Hz

BW: 500 k M G Hz

Generate Options | Filters

Options

ID	# Partial Options	# Chans	Partial Option 1 MCR	Partial Option 1 FCR	Partial Option 1 Channels	Partial Option 2 MCR	Partial Option 2 FCR	Partial Option 2 Channels	More Info
22	2	4	360 MHz	1.44 GHz	2	320 MHz	1.28 GHz	2	Show
1	2	5	184.32 MHz	1.47456 GHz	3	128 MHz	1.024 GHz	2	Show
6	2	5	184.32 MHz	1.47456 GHz	3	163.84 MHz	1.31072 GHz	2	Show
0	2	6	133.12 MHz	1.06496 GHz	4	125 MHz	1 GHz	2	Show
3	2	6	163.84 MHz	1.31072 GHz	2	150 MHz	1.2 GHz	4	Show
4	2	6	184.32 MHz	1.47456 GHz	3	153.6 MHz	1.2288 GHz	3	Show
5	2	6	184.32 MHz	1.47456 GHz	3	160 MHz	1.28 GHz	3	Show

Continue with the selected choice

Generate Partial Options

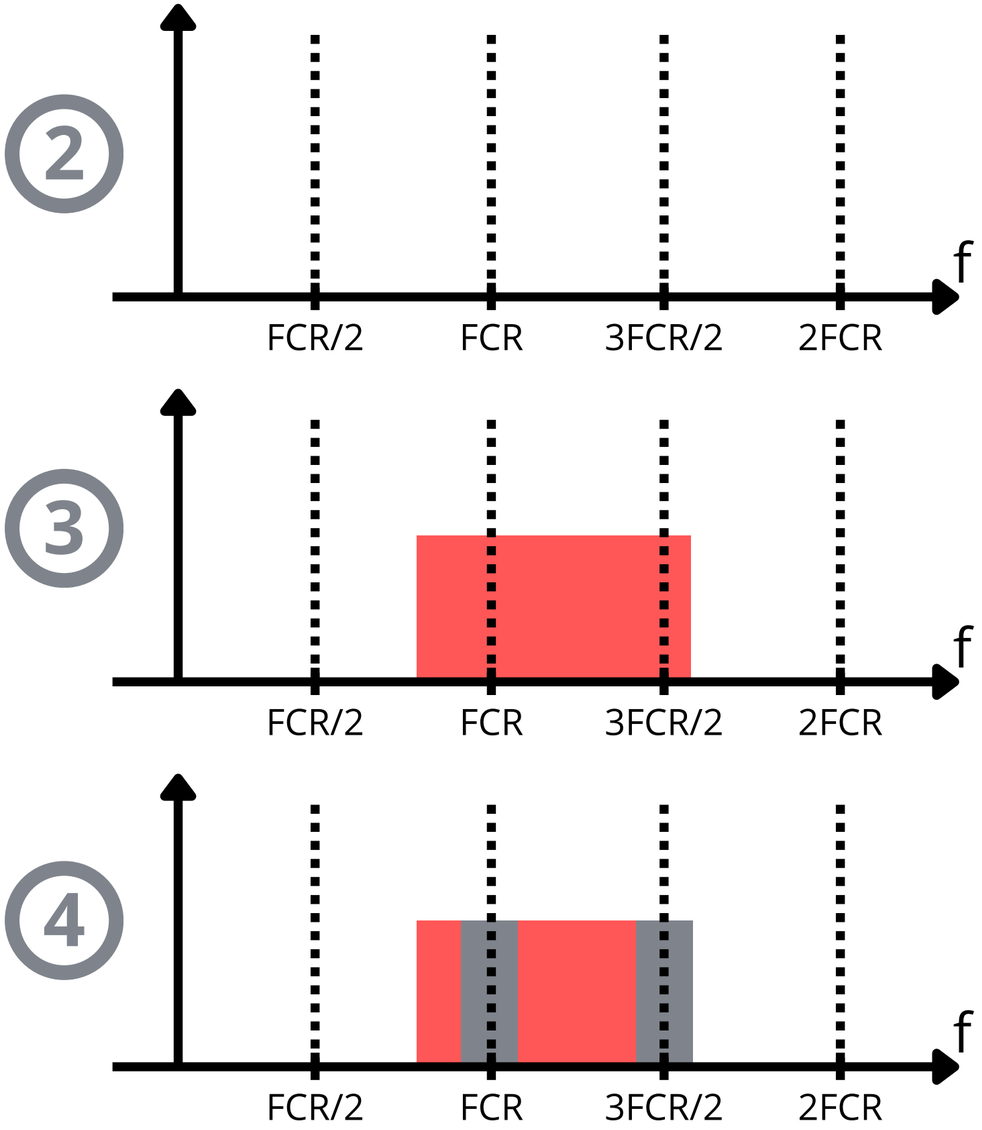
1) **Itera** per cada FCR de cada MCR

2) **Divideix** l'espectre segons el FCR
(Zones de Nyquist)

3) **Mira** a quines zones cau el nostre senyal

4) **Afegeix** marge a les Zones de Nyquist

5) **Emmagatzema** les dades a un .JSON



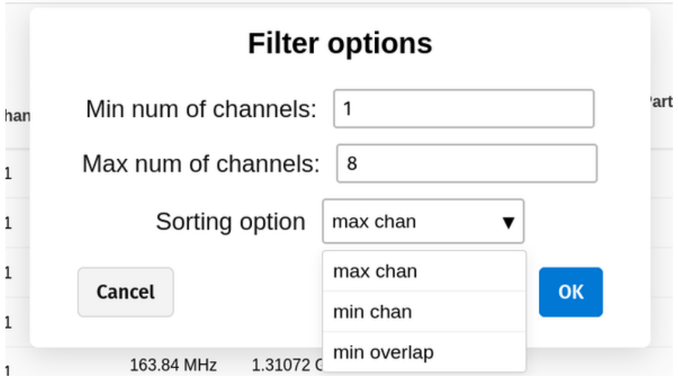
Generate Complete Options

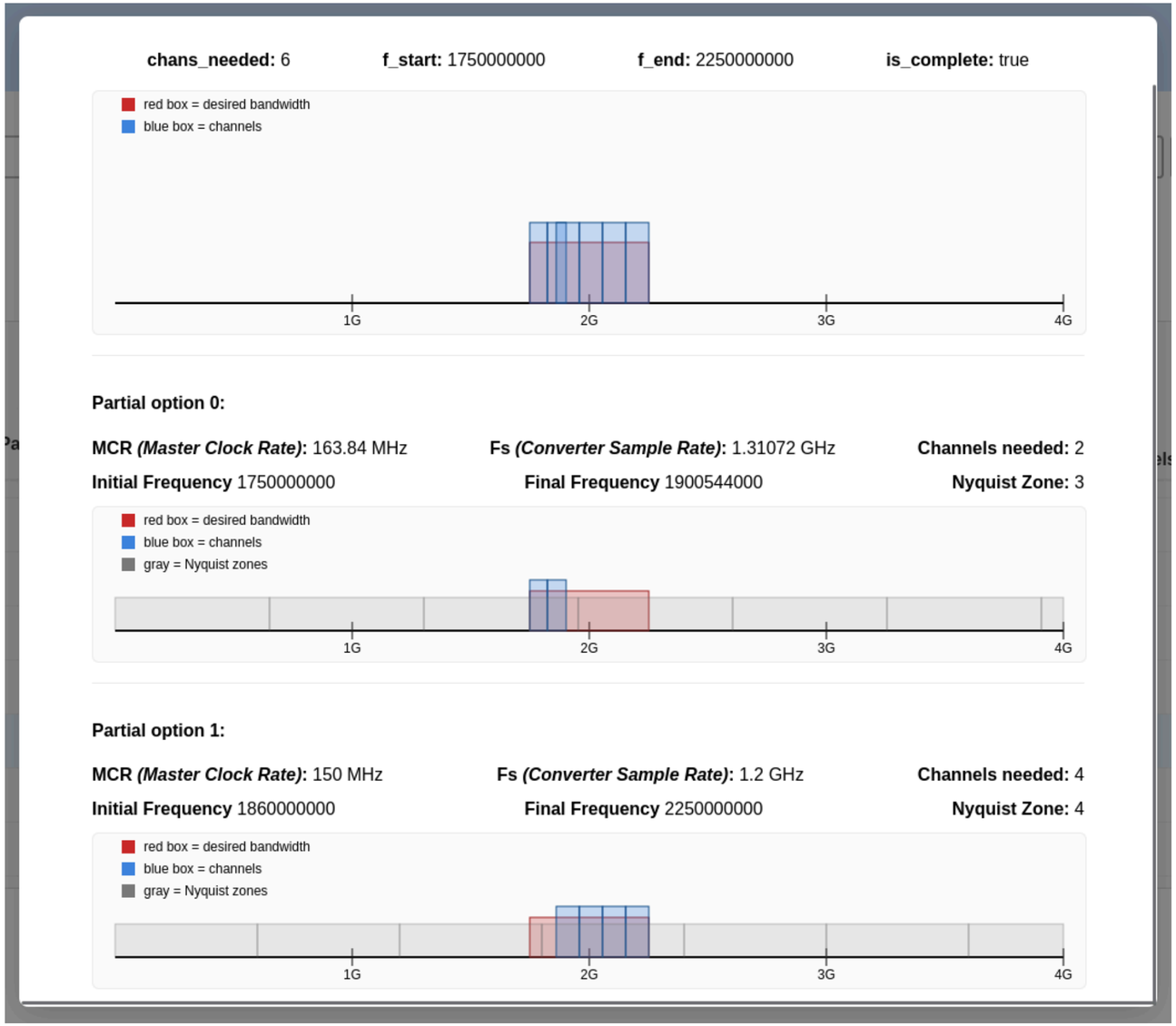
1) **Combina** totes les opcions parcials

2) **Mira** quines completen la banda sencera

3) **Emmagatzema** aquelles que generen opcions completes

4) **Filtra i Ordena** les opcions completes.





USRP X440 - wizard

F_c: 2 M G Hz

BW: 500 k M G Hz

Generate Options Filters

Options

ID	# Partial Options	# Chans	Partial Option 1 MCR	Partial Option 1 FCR	Partial Option 1 Channels	Partial Option 2 MCR	Partial Option 2 FCR	Partial Option 2 Channels	More Info
22	2	4	360 MHz	1.44 GHz	2	320 MHz	1.28 GHz	2	Show
1	2	5	184.32 MHz	1.47456 GHz	3	128 MHz	1.024 GHz	2	Show
6	2	5	184.32 MHz	1.47456 GHz	3	163.84 MHz	1.31072 GHz	2	Show
0	2	6	133.12 MHz	1.06496 GHz	4	125 MHz	1 GHz	2	Show
3	2	6	163.84 MHz	1.31072 GHz	2	150 MHz	1.2 GHz	4	Show
4	2	6	184.32 MHz	1.47456 GHz	3	153.6 MHz	1.2288 GHz	3	Show
5	2	6	184.32 MHz	1.47456 GHz	3	160 MHz	1.28 GHz	3	Show

Continue with the selected choice

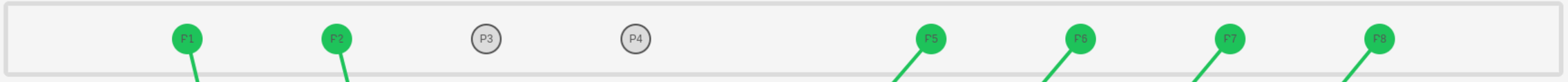
USRP X440 - wizard

Back

Hardware Connections

Ports

Port representation of the USRP x440:



Display of the channels for each of the partial options:

Option: 1

F_c: 1.788 GHz
BW: 75.272 MHz

F_c: 1.863 GHz
BW: 75.272 MHz

Option: 2

F_c: 1.909 GHz
BW: 97.500 MHz

F_c: 2.006 GHz
BW: 97.500 MHz

F_c: 2.104 GHz
BW: 97.500 MHz

F_c: 2.201 GHz
BW: 97.500 MHz

Connexions QSFP28

Connection:	IP Address:	Are Connected?	Is Connection Validated?
QSFP28_1	168.0.10.2	Yes	No
QSFP28_2	0.0.0.0	No	-

Validate Connections

Sample Rate

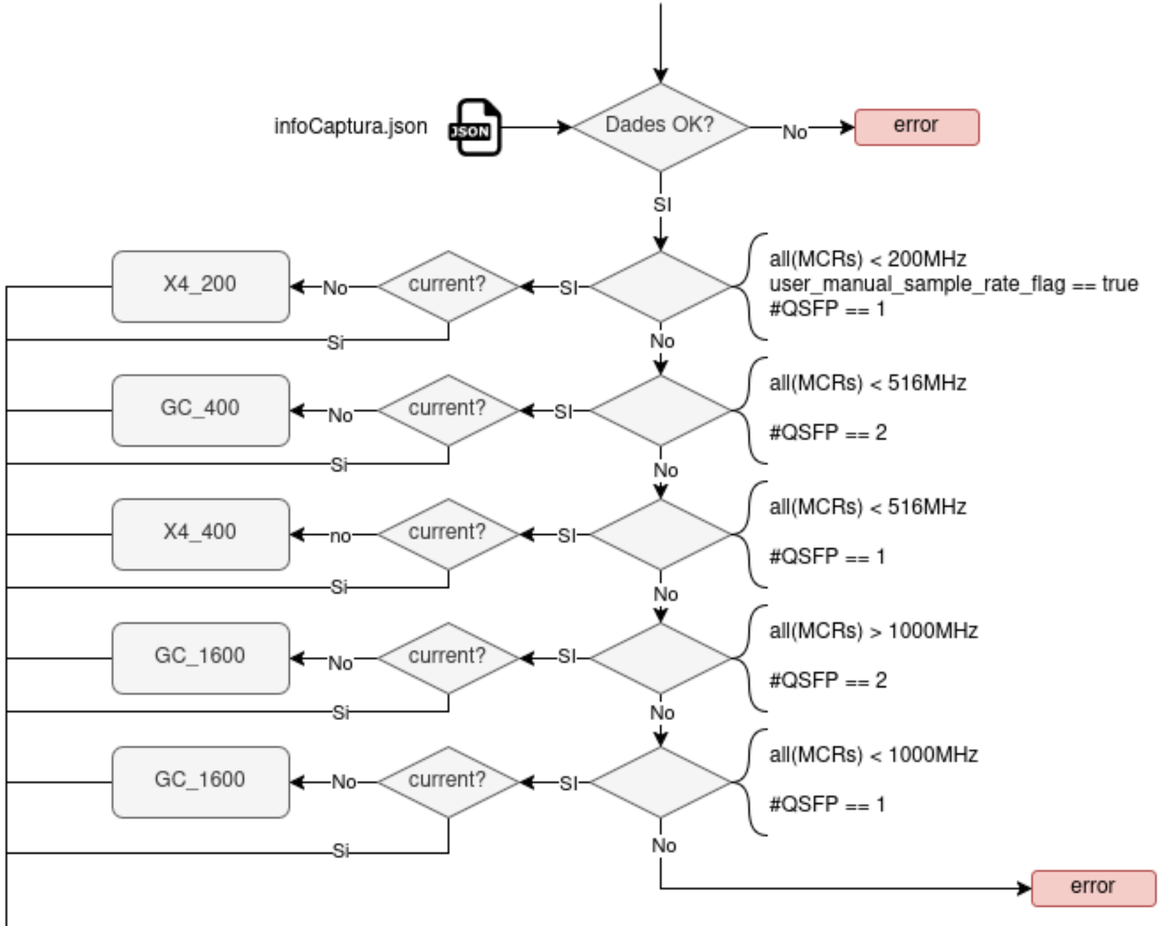
- [ ] Automatic
- [x] Manual 20e6

Capture Time (s): 3.5

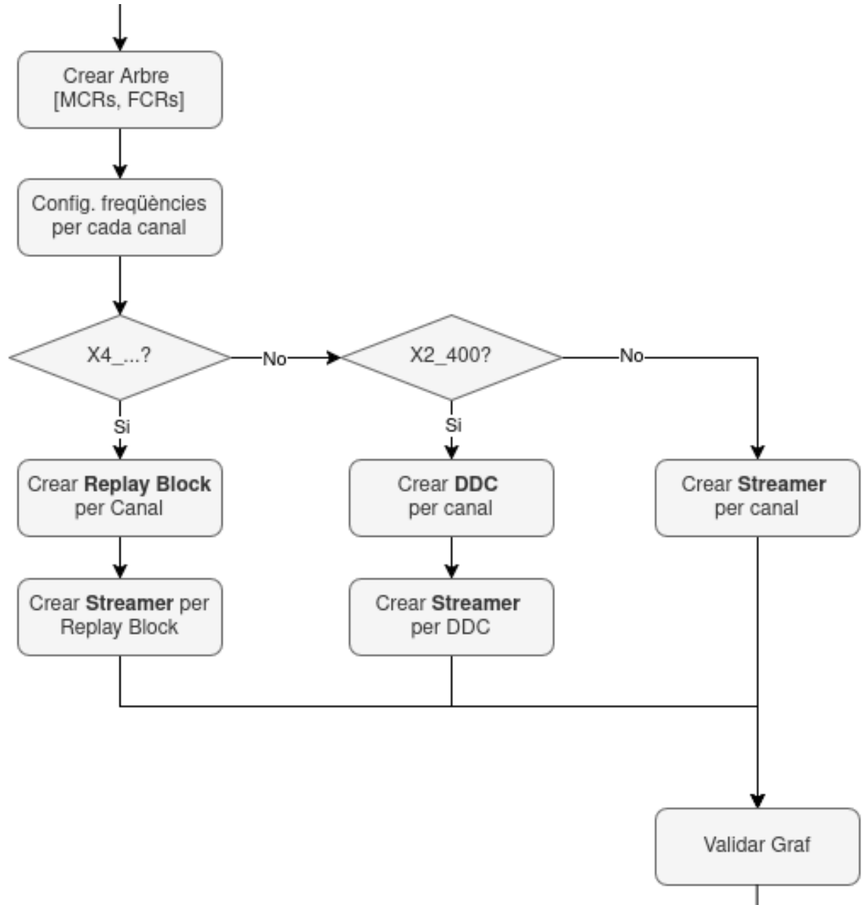
Capture

Script de Captura de Senyal

Validant dades i triant FPGA

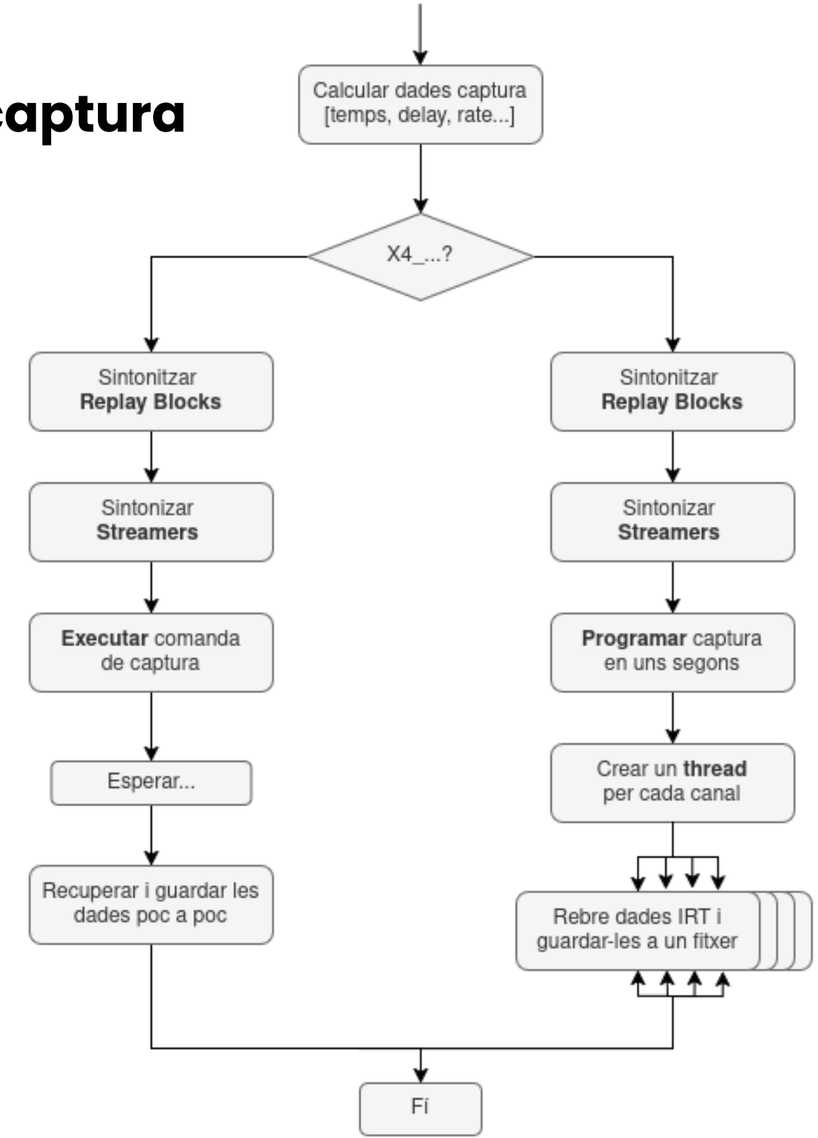


Creant el graf de captura



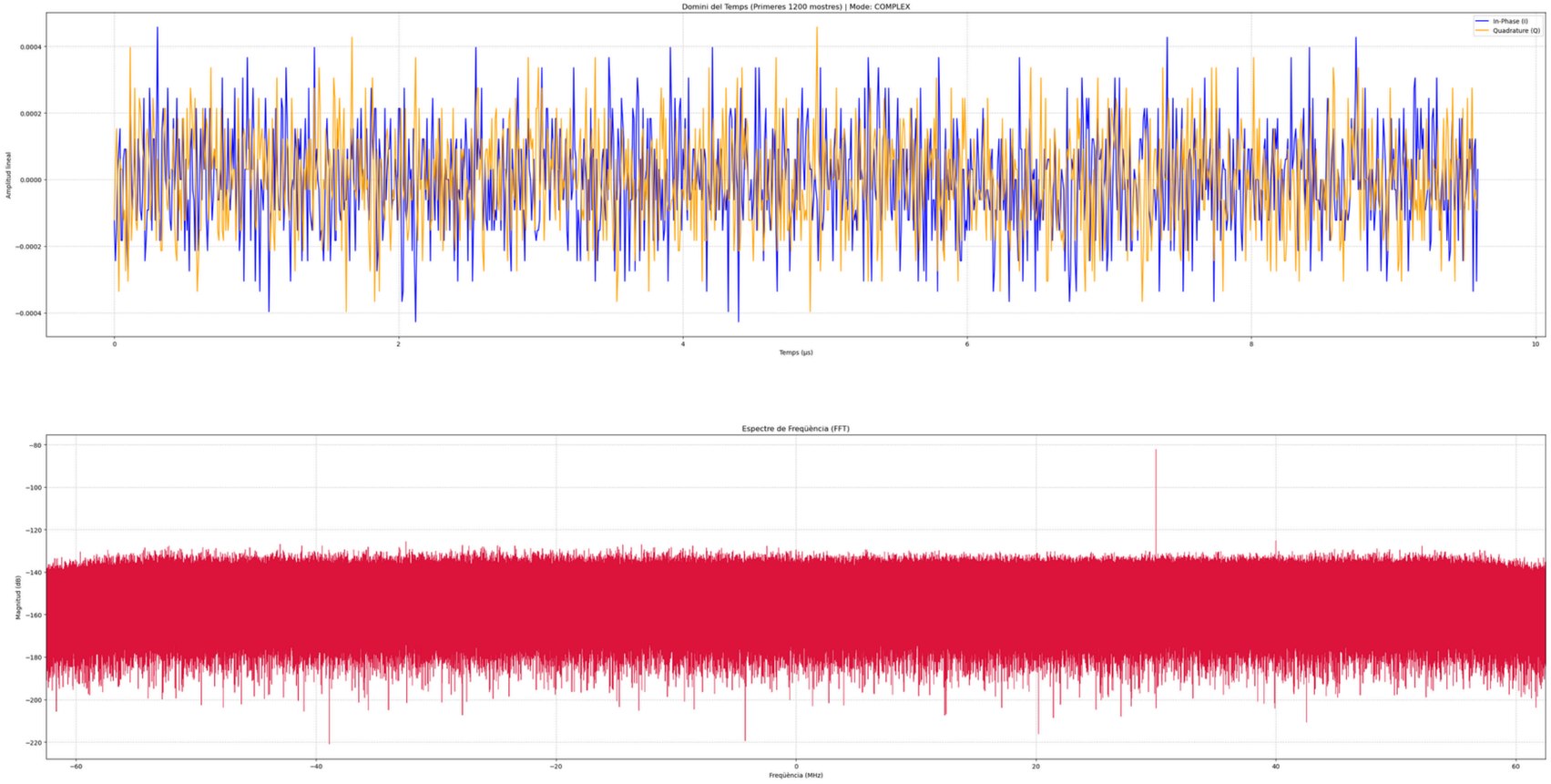
Script de Captura de Senyal

Executant la captura



Escenari 1: Captura de soroll

Valida el correcte funcionament (sense errors) del **programari** i la capacitat de **comunicació** amb la USRP.

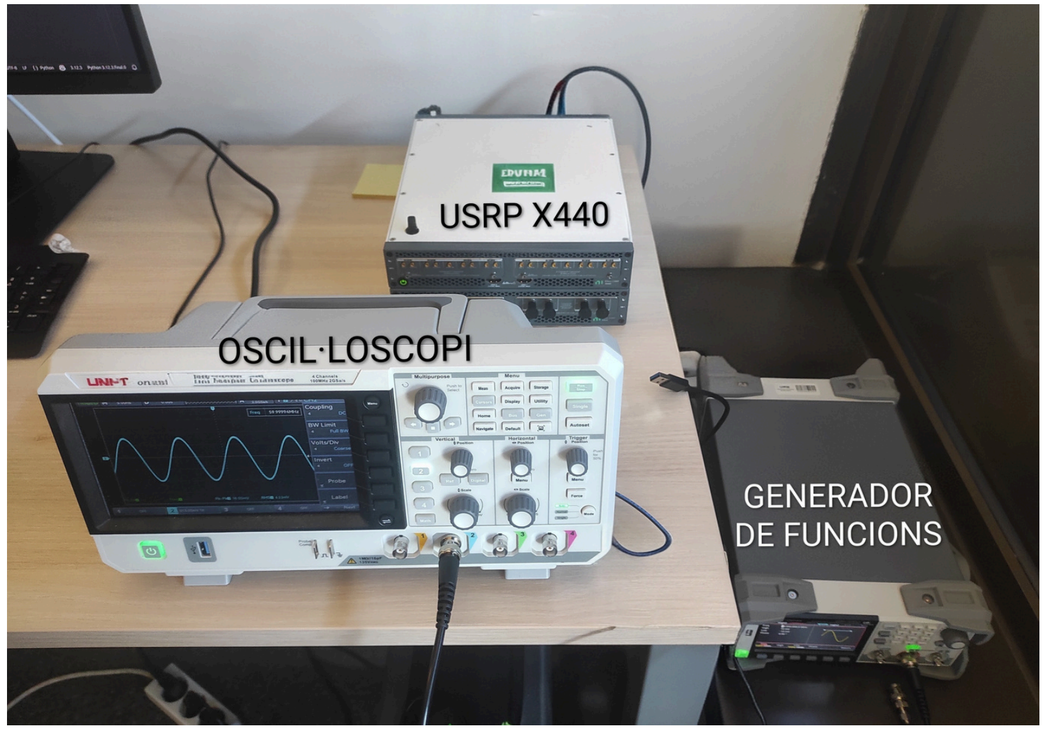


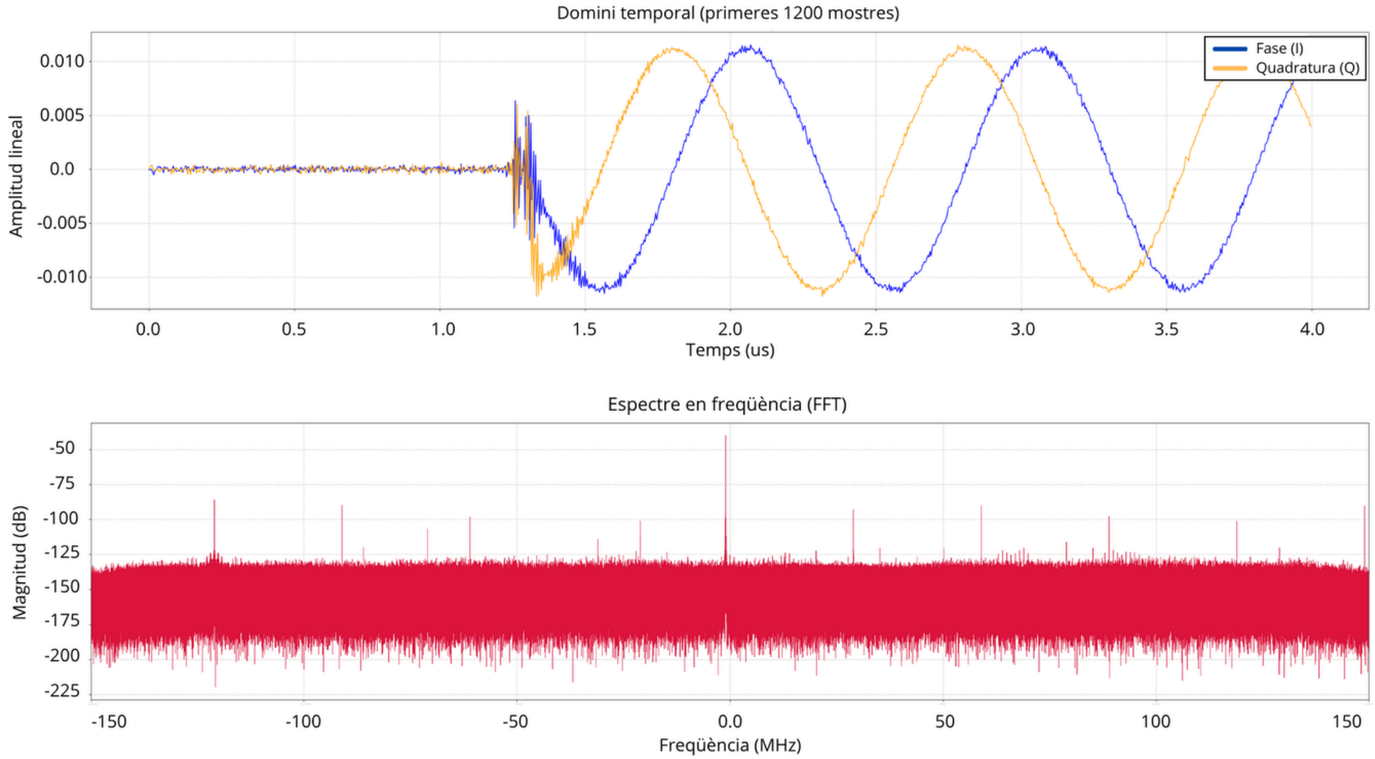
Escenari 2: Captura d'un senyal sintètic

Valida objectius:

1) **Estudi i comprensió** de l'arquitectura de Mostreig Directe i de la USRP X440.

2) **Creació** d'una **UI** i un **programari** que faciliten el procès de captura del senyal.

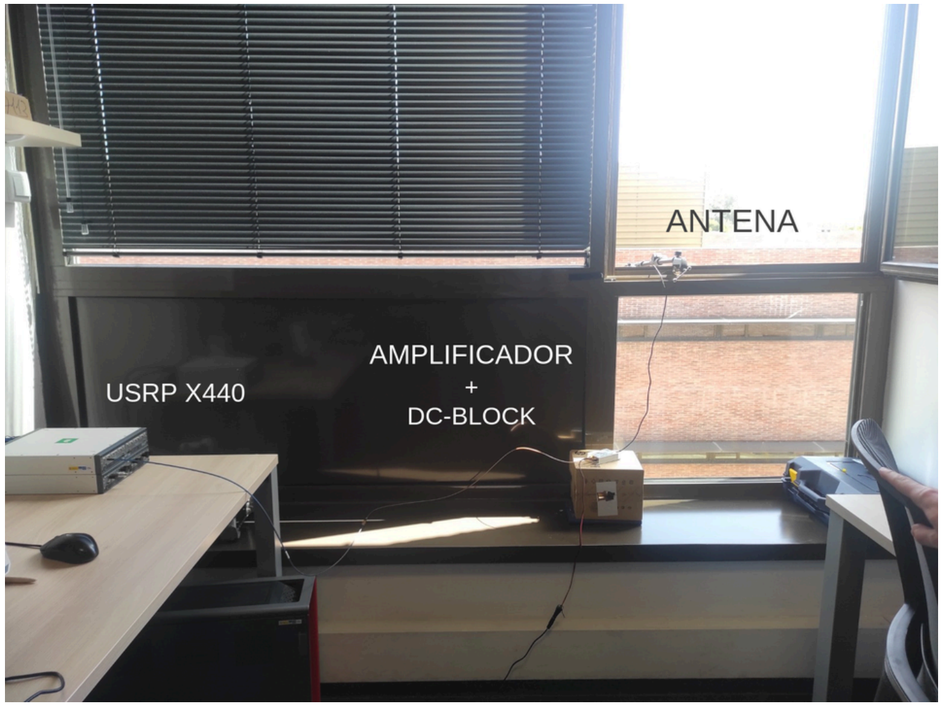


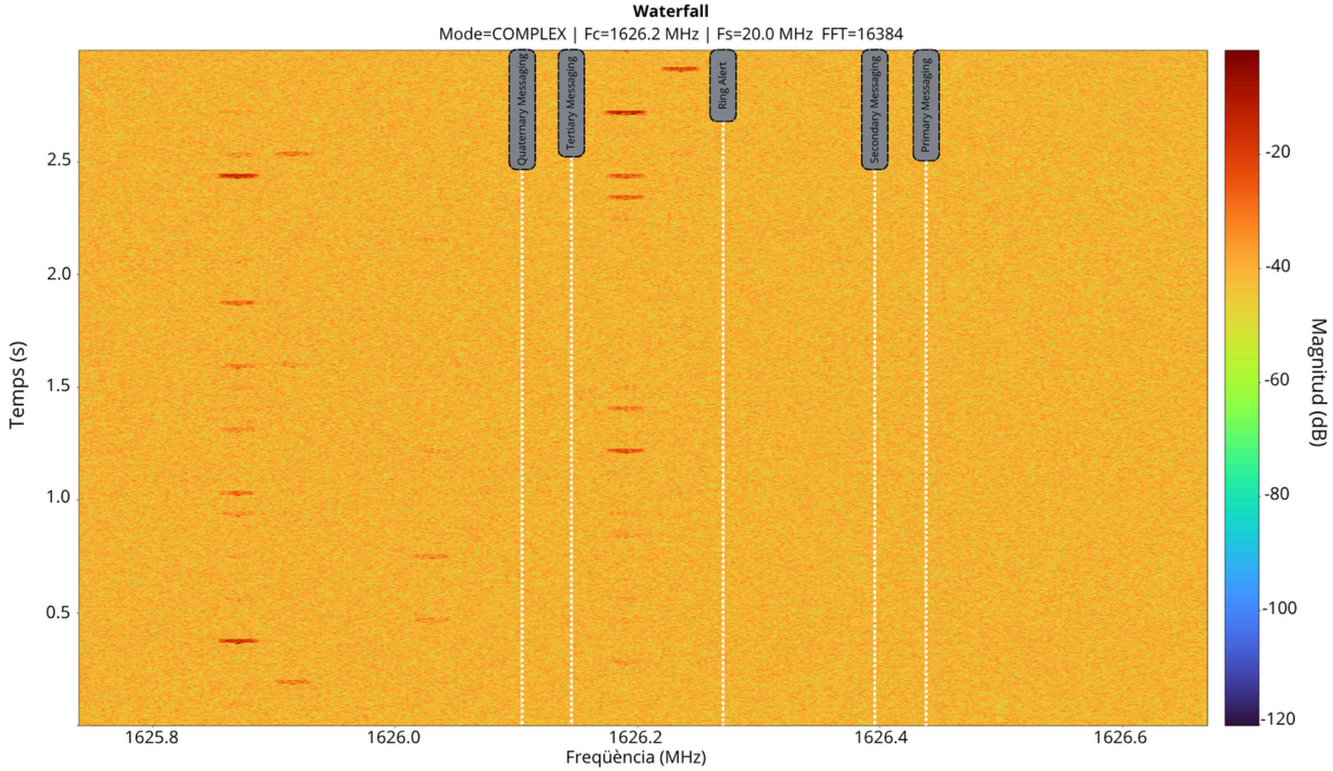


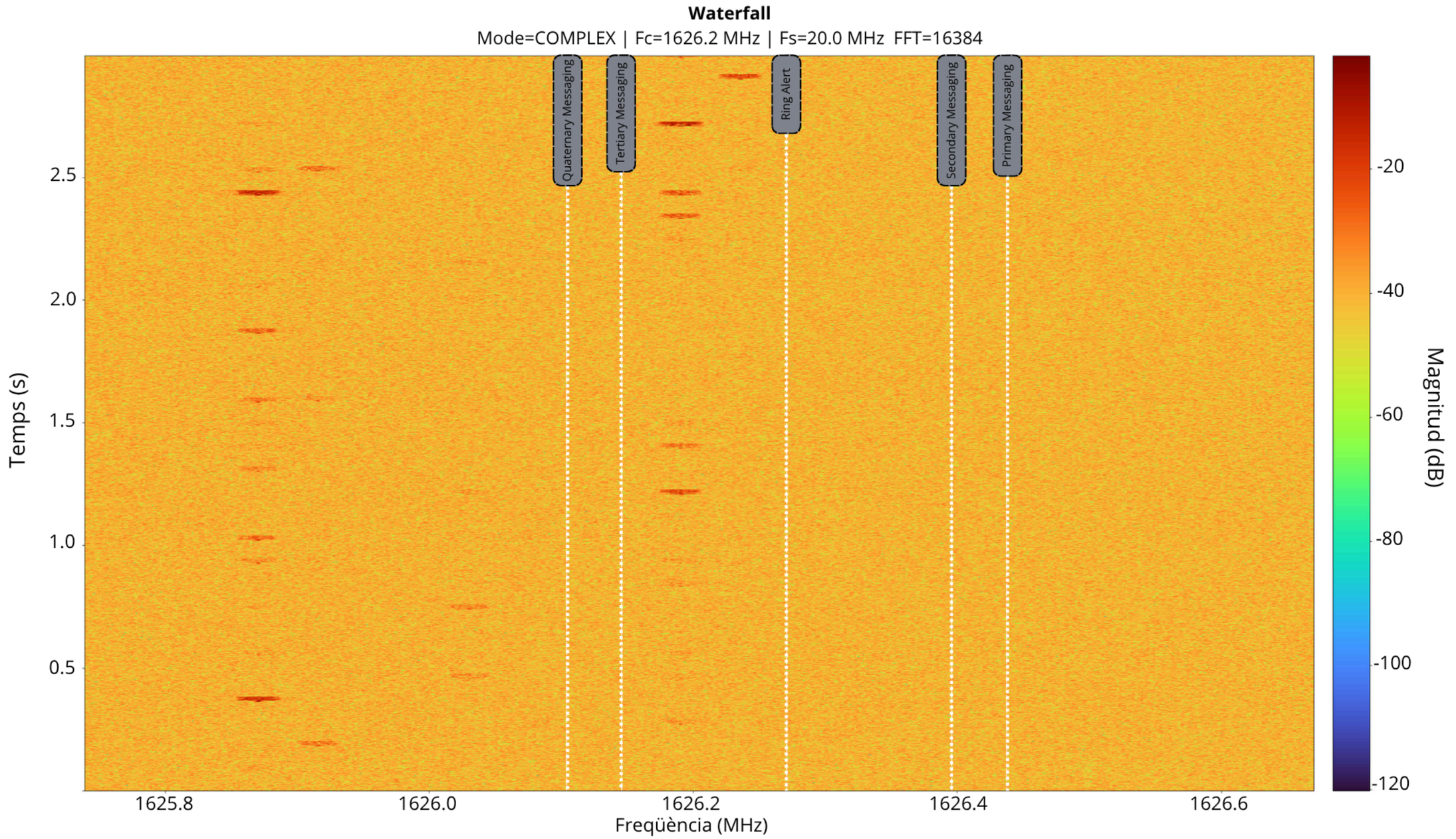
Escenari 3: Captura d'un senyal real

Valida objectius:

1) **Estudi i comprensió** de l'arquitectura de Mostreig Directe i de la USRP X440.
2) **Creació** d'una **UI** i un **programari** que simplifiquin el procés de captura del senyal.
3) Capacitat de **capturar un senyal real.**






Waterfall
Mode=COMPLEX | Fc=1626.2 MHz | Fs=20.0 MHz FFT=16384
Quaternary Messaging
Tertiary Messaging
Ring Alert
Secondary Messaging
Primary Messaging
Temps (s)
2.5
2.0
1.5
1.0
0.5
1625.8
1626.0
1626.2
1626.4
1626.6
Freqüència (MHz)
-20
-40
-60
-80
-100
-120
Magnitud (dB)

Objectius assolits:

- ☑ **Estudi i comprensió** de l'arquitectura de Mostreig Directe i de la USRP X440.
- ☑ **Creació** d'una **UI** i un **programari** que simplifiquen el procés de captura del senyal.
- ☑ Capacitat de **capturar un senyal real.** Que validi els punts anteriors.

Millores de cara a futures versions

Millora de la interfície per a fer més transparent el procès de captura.

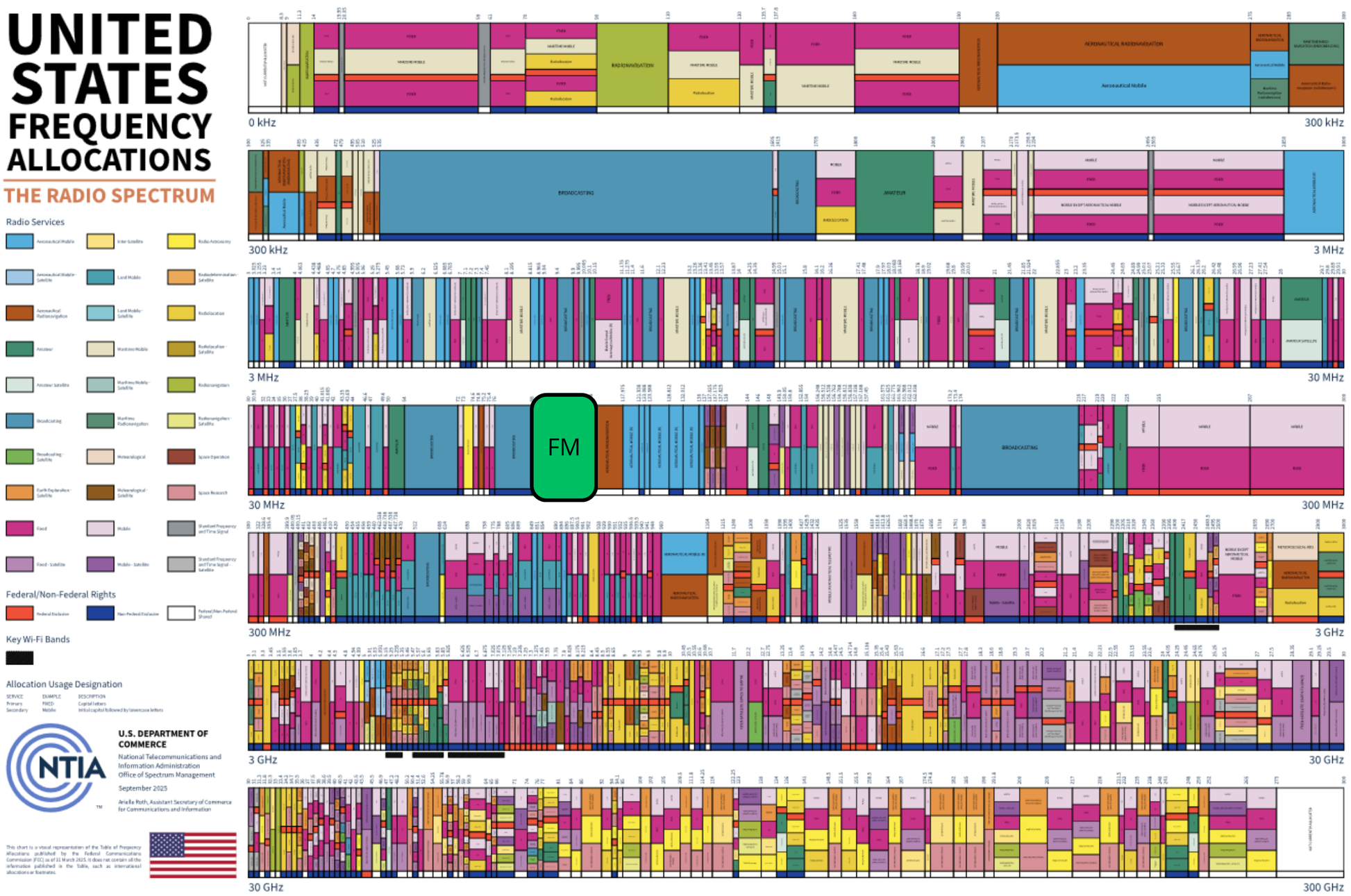
Provar de fer captures complexes de múltiples canals alhora per comprovar sintonització i captures dual rate.

Configurar el dispositiu i el servidor del laboratori per a crear una estació permanent

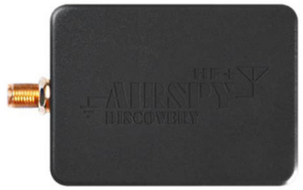
GRÀCIES PER LA VOSTRA ATENCIÓ

DIAPOS EXTRA

Aquesta diapo crec que la traurè



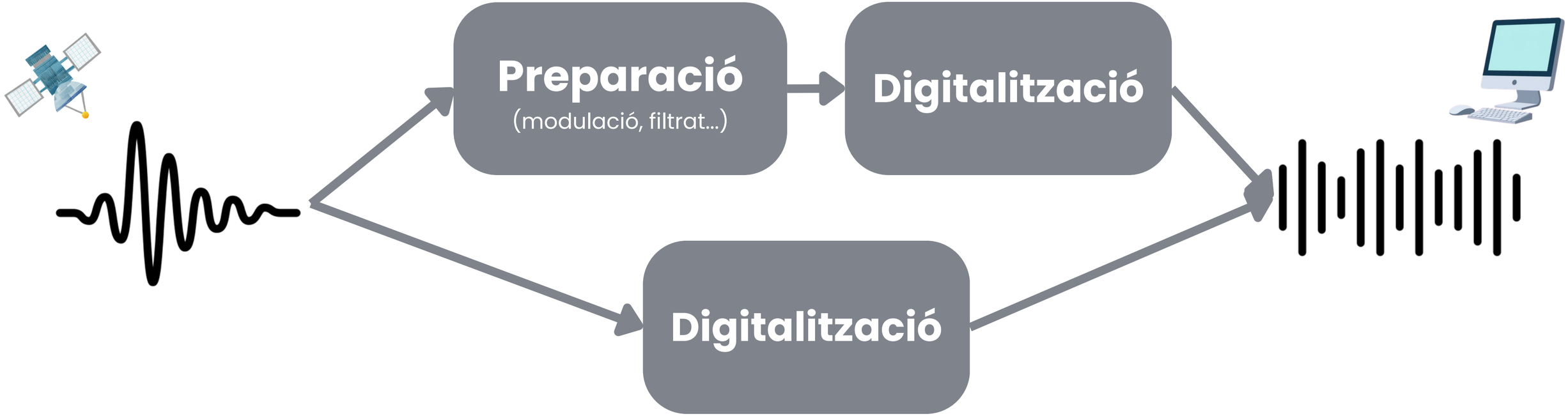
Software Defined Radios (SDRs)



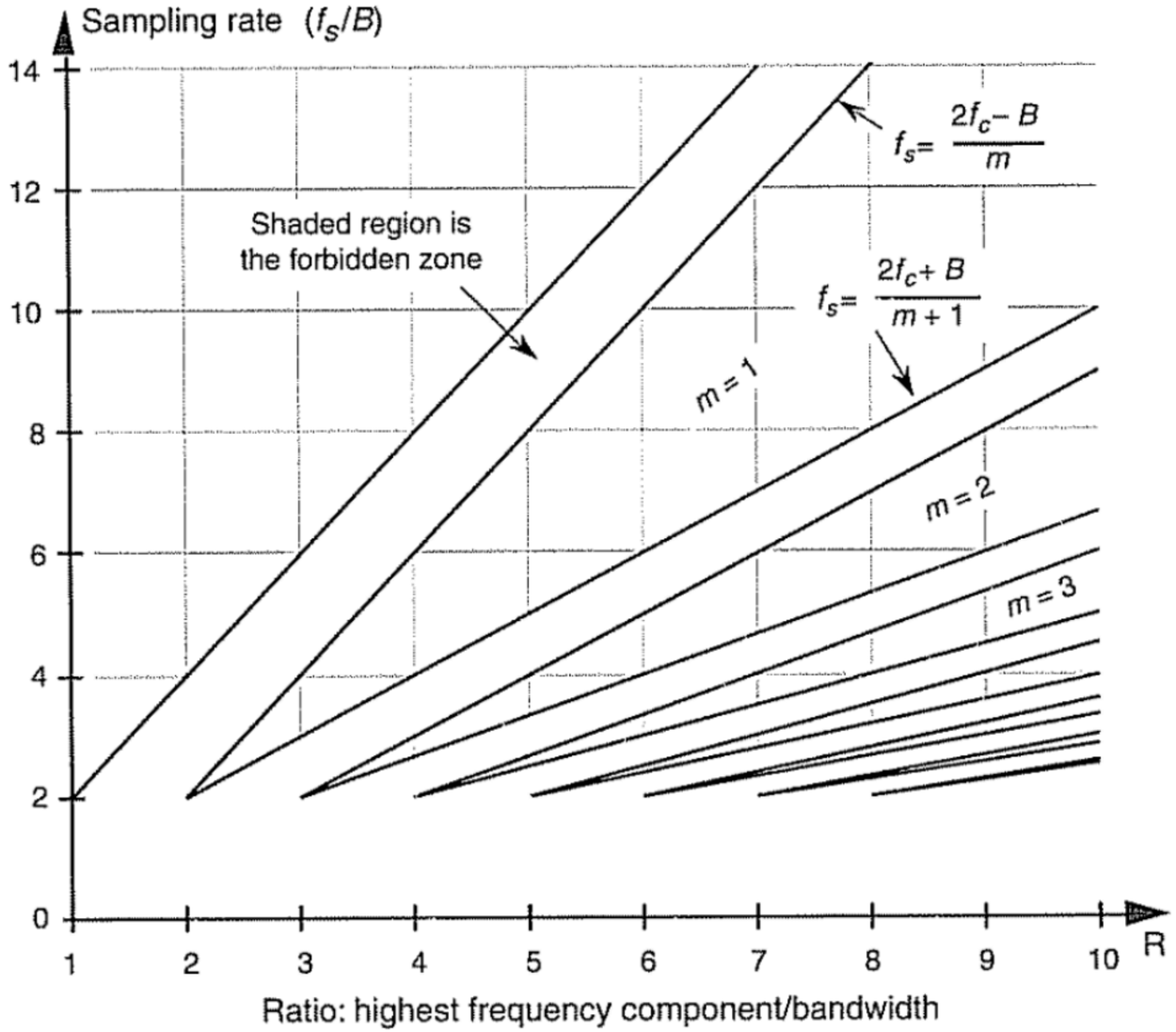
Model	Type	Frequency Range	Max Bandwidth	Bit Depth	Approx. Price
Standard FM Radio	RX Only	87.5 MHz – 108.0 MHz	∼200 kHz	N/A (Analog)	$10 – $30
RTL-SDR V4	RX Only	500 kHz – 1.7 GHz	2.4 MHz	8-bit	$35
Airspy HF+ Discovery	RX Only	0.5 kHz – 260 MHz	768 kHz	18-bit (DDC)	$170
HackRF One	TX/RX (Half)	1 MHz – 6 GHz	20 MHz	8-bit	$350
Adalm-Pluto (PlutoSDR)	TX/RX (Full)	325 MHz – 3.8 GHz	20 MHz	12-bit	$250
BladeRF 2.0 Micro xA4	TX/RX (Full)	47 MHz – 6 GHz	56 MHz	12-bit	$550

Arquitectures de receptors de radio (simplificat)

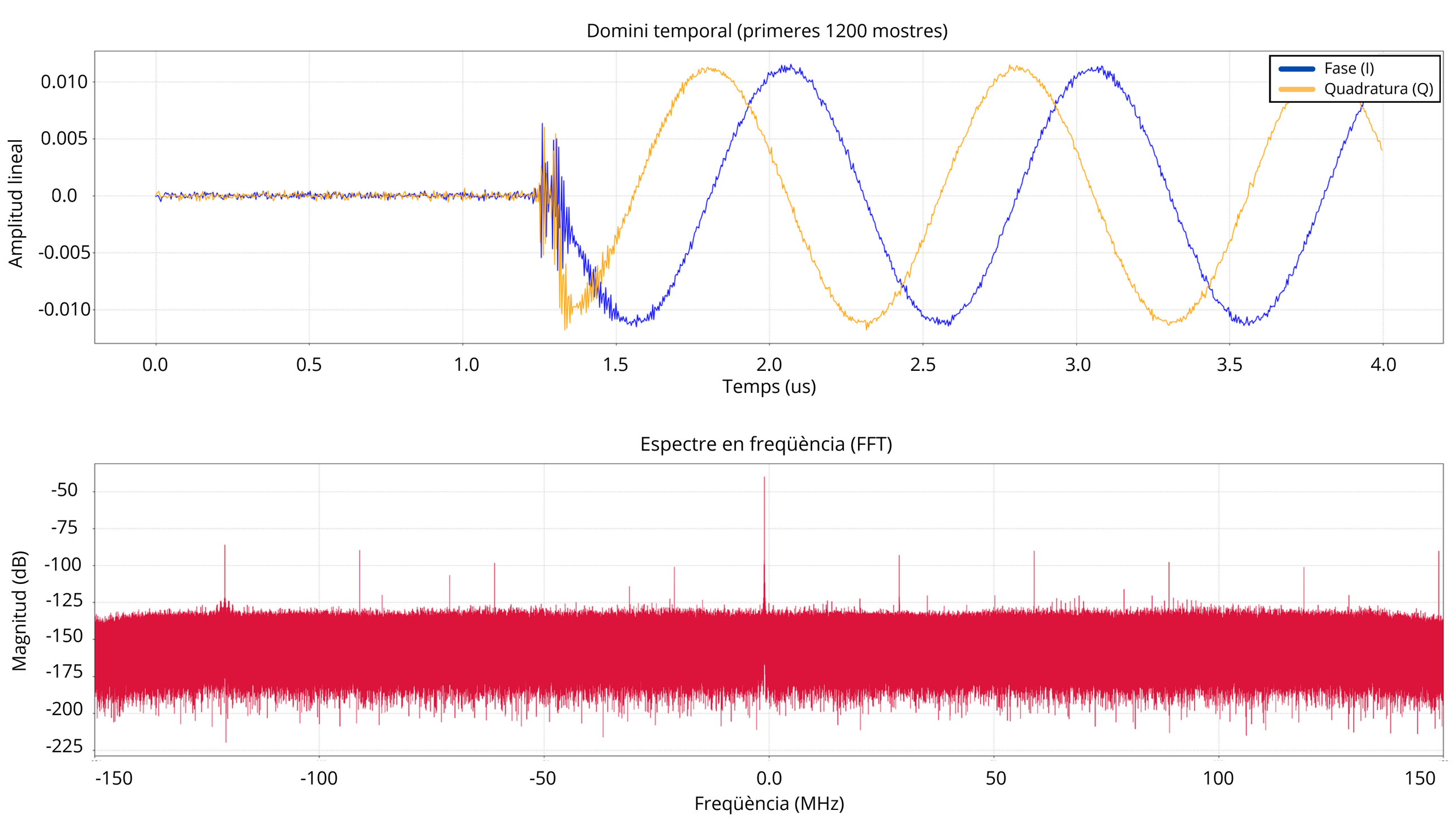
Superheterodina & Zero IF

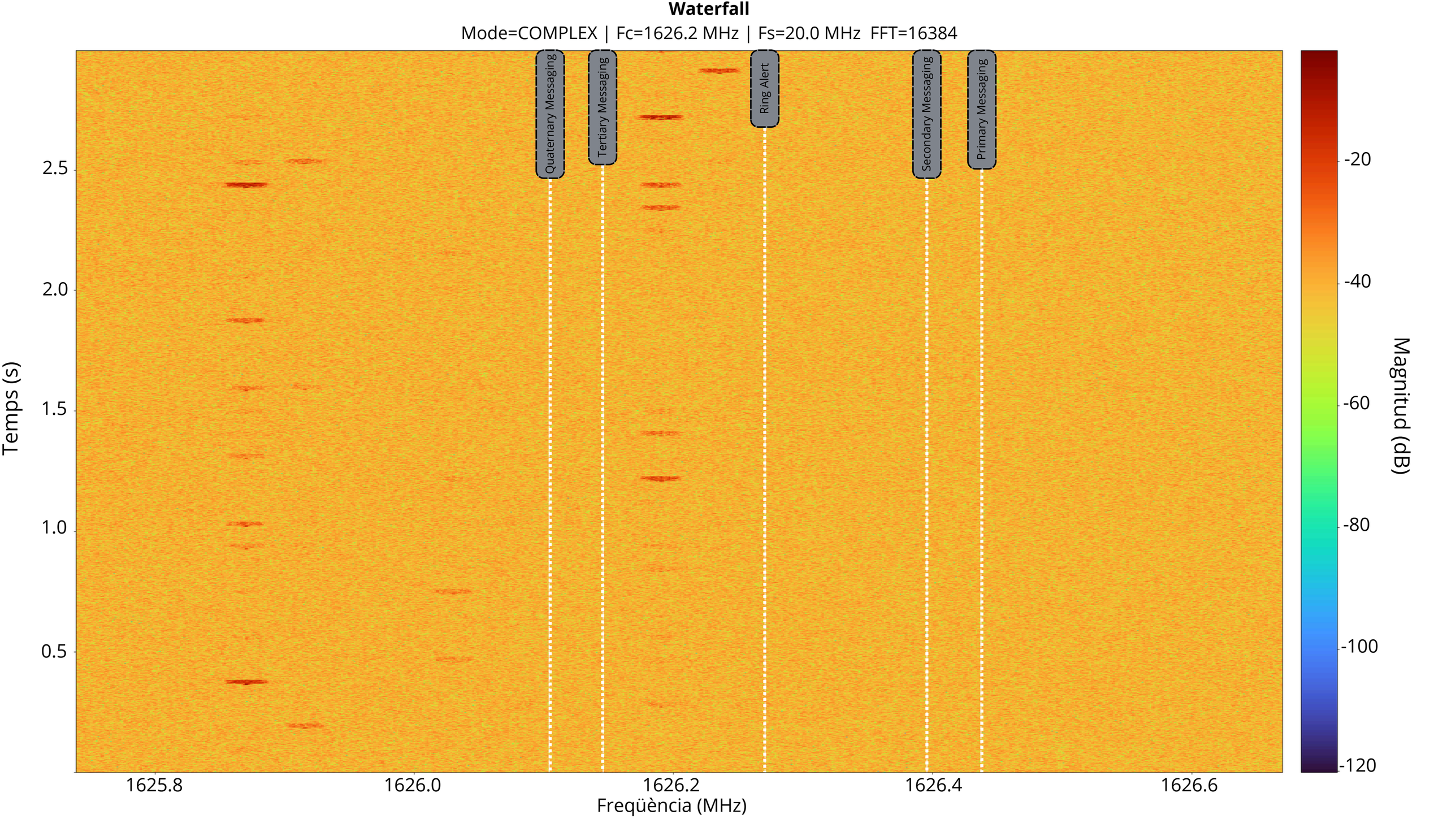


Mostreig Directe



Font: Richard G. Lyons. Understanding Digital Signal Processing. 3rd. Upper Saddle River, NJ: Prentice Hall, 2010. ISBN: 978-0137027415.


Domini temporal (primeres 1200 mostres)
Fase (I)
Quadratura (Q)
Amplitud lineal
0.010
0.005
0.0
-0.005
-0.010
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Temps (us)
Espectre en freqüència (FFT)
Magnitud (dB)
-50
-75
-100
-125
-150
-175
-200
-225
-150
-100
-50
0.0
50
100
150
Freqüència (MHz)


Waterfall
Mode=COMPLEX | Fc=1626.2 MHz | Fs=20.0 MHz FFT=16384
Quaternary Messaging
Tertiary Messaging
Ring Alert
Secondary Messaging
Primary Messaging
Temps (s)
2.5
2.0
1.5
1.0
0.5
1625.8
1626.0
1626.2
1626.4
1626.6
Freqüència (MHz)
-20
-40
-60
-80
-100
-120
Magnitud (dB)